

Member ID: _____

Time: _____

Rank: _____



SYSTEM ADMINISTRATION USING MICROSOFT® (310)

NATIONAL 2022

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each)

_____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice: Identify the letter of the choice that *best* completes the statement or answers the question.

1. Which of the following commands can be used to resolve the error: "The trust relationship between this workstation and the primary domain failed"?
 - a. Reset-ComputerMachinePassword.
 - b. Nltest /sc_reset_machineaccount
 - c. New-DomainWorkstationRequest
 - d. User-Reset-ComputerDomain
2. Which additional role is automatically added when installing the ADDS role?
 - a. DHCP
 - b. DNS
 - c. DHA
 - d. IPSEC
3. Which of the following is *not* a common cause of the error: "There are currently no logon servers available to service the logon request."?
 - a. An invalid DNS server is configured
 - b. The domain controller is offline
 - c. An incorrect DNS record is being returned for the domain
 - d. The local machine is using a VPN
4. Your network contains an Active Directory domain named contoso.com. The domain contains a server named Server1 that runs Windows Server 2019. You create an inbound rule by using Windows Firewall with Advanced Security. You need to configure the rule to allow Server1 to accept unsolicited inbound packets. Which setting in the rule should you configure?
 - a. Interface types
 - b. Authorized computers
 - c. Remote IP address
 - d. Edge traversal
5. What makes a global catalog different from other domain controllers?
 - a. Contains read-only partitions exclusively
 - b. Used for indexing legacy schemas in other forests
 - c. Contains read-only copies of all domains in the forest
 - d. Used for authentication of only the parent domain in the forest
6. Which role would you install for managing printer resources in a domain?
 - a. Print and document services
 - b. File and printer services
 - c. Network printing administration
 - d. Printers and faxes console

7. What is the difference in a striped volume and a RAID 5 volume?
 - a. Striped volumes contain identical copies of all the data on each members
 - b. Striped volumes will fill space on member drives consecutively
 - c. Both use stripes but RAID 5 provides redundancy through parity
 - d. RAID 5 is only available through a hardware controller
8. What is storage spaces?
 - a. Microsoft's software defined storage feature
 - b. An optional Sys Internals tool for disk analysis
 - c. A cloud only file repository available through Azure
 - d. An industry term for external drives
9. What is the minimum number of disks that must be used in a storage pool?
 - a. Two of any matching type
 - b. Three of any type
 - c. Four with at least 2 SSDs
 - d. Five SAS or SATA drives
10. Which of the following can be accomplished with the Test-NetConnection PowerShell command?
 - a. DNS propagation
 - b. Updating DNS settings
 - c. Route Tracing
 - d. DNS flushing
11. Which RAID level is described by a JBOD configuration?
 - a. RAID 0
 - b. RAID 1
 - c. RAID 5
 - d. RAID does not use JBOD configuration
12. What is the purpose of the SNMP protocol?
 - a. SNMP is used by devices within a network use to communicate problems with data transmission
 - b. SNMP is used to transfer e-mail messages and attachments
 - c. SNMP is used to monitor the network, detect network faults, and sometimes even used to configure remote devices
 - d. SNMP is used to provide a mechanism to transport PPP traffic through an SSL/TLS channel
13. On Windows Server 2019 using a standard installation of DNS where is the default file location for the DNS database?
 - a. %systemroot%\System32\DNS
 - b. %systemroot%\System32\DNS\dnsbackup.txt
 - c. %systemroot%\System32\DNS\data\dnsconfig
 - d. %systemroot%\System32\dns\dns.cfg

14. How many OS instances can be virtualized under a Windows server 2019 standard license?
- a. Standard does not contain virtualization rights
 - b. Two
 - c. Four
 - d. Unlimited
15. How many OS instances can be virtualized under a Windows server 2019 datacenter license?
- a. Datacenter does not contain virtualization rights
 - b. Two
 - c. Four
 - d. Unlimited
16. What is the primary benefit of DNS split tunneling?
- a. Improve reliability by removing single points of failure
 - b. Load balancing of domain authentication requests for remote users
 - c. Reduce congestion on VPN connections by directing some traffic to public resources
 - d. Less CPU utilization on network appliances
17. Which of the following methods would grant an Active Directory user permission to connect over VPN?
- a. Launch a command prompt with admin rights and use the command “netsh ras set user <username> permit”
 - b. Use active directory users and computers and select deny access on the dial-in tab of the accounts properties
 - c. Launch a command prompt with admin rights and use the command “netsh ras set user <username> deny
 - d. Launch a command prompt with admin rights and use the command “net ras set user <username> permit”
18. What is the purpose of a DHCP reservation?
- a. A permanent IP address assignment to a specific client based on MAC
 - b. Prevent a range of specified addresses in a range from being assigned
 - c. A valid range of IP addresses available for assignment to clients
 - d. Reserve specific DHCP links for use by preferred clients
19. Users in your active directory domain report they are losing internet access when logged into the VPN but can access domain resources normally. Which of the following is a possible solution?
- a. The boundary router needs to be replaced
 - b. Flush the DNS server cache and ensure a forwarder is configured
 - c. Use Routing and Remote Access to expand the VPN’s IP address pool
 - d. Instruct users to contact their ISP for assistance unblocking their home modems

20. What is the primary function of a delegation name (DNAME) DNS record?
- It contains the address of an e-mail server responsible for the domain
 - It is used for reverse DNS lookups
 - It identifies the name server that is authoritative for the domain
 - It is an alias for one or more A records with different names
21. When a device receives the DHCPINFORM packet what sort of information might be contained in the options fields?
- DNS server, domain name
 - Client new address, subnet mask
 - DNS server, domain name, client new address, subnet mask
 - Only the subnet mask
22. Which of the following methods can be used to transfer FSMO roles between domain controllers?
- Set-ADDirectoryServerOperationMasterRole <destination> 0,1,2,3,4
 - Ntdsutil Transfer FSMO -all <destination>
 - Set-ADDirectoryServerOperationMasterRole <destination> 0,1,2,3,4 or Ntdsutil Transfer FSMO -all <destination>
 - None of the other options are correct
23. Which of the following accurately describes WMI ?
- An MMC snap-in for management of user desktops
 - A firewall-friendly protocol that allows hardware and operating systems, from different vendors, to interoperate
 - An infrastructure protocol for management data and operations on Windows-based operating systems
 - An Internet standard protocol used by email clients to retrieve email messages
24. You want to implement a new server role in Windows 2019 that allows you to create and enforce organization-wide network access policies for connection request authentication and authorization. What role is the best choice for this task?
- Remote access services
 - Network policy and access services
 - Network controller services
 - Directory authentication services
25. Which if the following is true regarding Active Directory Forests?
- Trees may contain Parent-Child trusts but not Tree-Root trusts
 - Every tree shares a namespace with the forest root domain
 - A forest is always a dingle DNS name space
 - None of the other options are correct.

26. Which boot option allows you to access advanced system recovery and restoration tools?
- Active directory restore mode
 - Safe mode with networking
 - Advanced recovery mode
 - Repair your computer
27. Which group policy option limits active hours a user may be logged in?
- Account policy: Disallow user sessions when logon hours expire
 - Account policy: Log off users when logon hours expire
 - Network security: Enforce logon hours
 - Network security: Force logoff when logon hours expire
28. Which of the following Resource manager tabs will display threads for active processes?
- Network
 - CPU
 - Memory
 - Disk
29. Which of the following is commonly used to manage a Windows Server 2016 Core installation using a GUI interface?
- RSAT
 - PowerShell
 - PuTTY
 - VMMSC
30. What special character can be placed at the end of a share name to hide it from the network browser?
- !
 - \$
 - ?
 - *
31. What Active Directory containers are used for organizing and managing objects in a domain?
- Organizational units
 - Universal groups
 - Computer accounts
 - Security principals
32. Which of the following statements regarding the conversion of group scope is true?
- Universal groups that are members of other universal groups can be converted to domain local groups
 - Security groups can be converted to global groups
 - Empty domain local groups can be converted to universal groups
 - Global groups can't be converted to universal groups

33. You are an administrator in a multi-domain environment and have been asked to improve response times for resolving DNS queries for a non-Active Directory domain from your Active Directory Domain. What type of DNS zone can be used decrease query times with minimal overhead?
- A forward zone
 - A reverse zone
 - A stub zone
 - A glue zone
34. When Active Directory replication is failing what log is most useful to determine the root cause of the failure?
- DFS replication log
 - Active directory replication log
 - SYSVOL transfer log
 - Directory service log
35. When using differential backups what is a potential negative impact of this configuration?
- You need additional space on the backup target
 - Write performance on the source volume is decreased
 - It takes longer to perform the backup
 - It takes longer to perform the restore
36. What command can be used to refresh the group policy remotely on a Windows 10 client?
- Invoke-GPUdate
 - Gpupdate.exe
 - Remote-GPUdate
 - Gpremove.exe
37. What PowerShell cmdlet is used to set permissions on one or many Group Policy Objects?
- Chmod-GPO
 - Change-GPOSecurity
 - Set-GPPermission
 - Set-GPOSecurity
38. A network user calls into the helpdesk after trying to login to their workstation several times to have their password reset. You reset the user's account password, and the user is still unable to login. What is the most likely reason?
- The user is in a new OU and lacks log on permissions to the computer used
 - The user's account has been locked due to frequent failed password attempts
 - The computer is having issues communicating with the domain and is attempting to use cached credentials
 - The global catalog server has not replicated the user's account information to the rest of the domain

39. When multiple Group Policies Objects are applied what order determines the resultant set of policy?
- Site-linked GPOs, local policies, domain-linked GPOs, OU-linked GPOs
 - Domain-linked GPOs, OU-linked GPOs, local policies, site-linked GPOs
 - Site-linked GPOs, domain-linked GPOs, OU-linked GPOs, local policies
 - Local policies, site-linked GPOs, domain-linked GPOs, OU-linked GPOs
40. Which of the following Internet Information Services (IIS) built-in groups allows anonymous access to web resources?
- Nobody
 - IIS_USERS
 - Anonymous logon
 - IIS_ANON
41. What performance forward RAID level features no redundancy with disk striping?
- RAID 0
 - RAID 1
 - RAID 5
 - RAID 10
42. What Group Policy Object determines the number of unique new passwords associated with a user account before an old password can be reused?
- Maximum password age
 - Minimum password age
 - Passwords must meet complexity requirements
 - Enforce password history
43. What is a possible downside of using full event auditing policies?
- Event auditing can provide an attacker with valuable information regarding the structure of a network's security policies
 - Event auditing can fill the Security event log and consume considerable resources
 - Event auditing can trigger access warning messages for users
 - Event auditing can affect the access control list security for protected objects
44. How much time will expire on a DHCP client's lease before it sends a lease renewal request to the DHCP server it obtained it from?
- 50% of lease interval
 - 55.5% of lease interval
 - 80% of lease interval
 - 87.5% of lease interval
45. The Windows server backup tool can only use NTFS volumes.
- True
 - False

46. You need to use the system file check utility to restore a corrupted Windows file. Where does the system file check get the uncorrupted version from?
- The C:\Windows\System32 folder
 - The C:\Windows Drivers folder
 - The C:\Windows\hidden\sysfiles folder
 - The C:\Windows\System\backup folder
47. Which of the following is the correct format of DNS reverse lookup zone for the network 10.50.12.0/24?
- 10.50.12.arpa
 - 10.50.12.in-addr.arpa
 - 12.50.10.in-addr.arpa
 - 12.50.10.arpa
48. Which of the following advanced boot options will allow you to load registry settings from the previous login on a Windows 10 client?
- Last know good configuration
 - System restore
 - Debugging mode
 - Safe mode with command prompt
49. An IP reservation is an association between an IP address and what type of client identifier in DHCP?
- NIC GUID
 - MAC address
 - User class
 - Vendor class
50. What is the difference in DNS records of the A and AAAA type?
- A is used for IPv4 addresses while AAAA is used for IPv6
 - AAAA records are used for servers with a higher priority
 - A records are no longer used
 - AAAA records are used to refer to resources in another domain